

DECODE SIH 2026

ResQNet — Disaster Response Intelligence & Coordination Platform

PRESENTED
BY: DARK DEV

TEAM MEMBERS

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PROBLEM STATEMENT

Problem Title :

Disaster Response Intelligence Platform for Flood Prediction, Emergency Planning, and Resource Allocation

What does this Problem Aim to Solve?

During disasters like floods, many people need help at the same time, but information is often scattered between citizens, volunteers and emergency teams. Poor connectivity, limited resources and manual decisions can slow down the response. This problem aims to organize incident information, identify priorities and help teams coordinate the right people and resources.

Why did your team choose this?

We chose this problem because disasters require quick decisions and proper coordination. We wanted to build something that can actually help during emergencies instead of only displaying information. Our idea combines AI, offline-first reporting, location-based information, and volunteer and resource management to help emergency teams respond more effectively.

REAL-WORLD PROBLEM ALIGNMENT

Who faces this issue? (E.G., Rural Students, Local Governance, Citizens, etc.)

- **Citizens in disaster-affected areas**, especially those who need urgent help, evacuation, medical support or essential resources. **Volunteers, responders and local authorities** also face difficulties in coordinating help for these people during large-scale emergencies.

What's the current gap in the system?

- Disaster information is often scattered across **calls, messages and different channels**. During emergencies, unreliable connectivity and limited resources make it difficult to know **which incidents are most urgent, where help is needed, and who should respond**.

How does your proposed solution fill that gap practically?

- **ResQNet brings reporting, incident prioritization, location information, volunteer coordination and resource management into one platform**. Its offline-first reporting helps capture important information even during connectivity problems, and the information can be synchronized when the connection returns.

PROPOSED SOLUTION

What are you Building?

ResQNet is a disaster response platform that connects citizens, emergency operators, volunteers and resources through a unified workflow. Citizens can report incidents with important details and location information, while the offline-first approach allows critical reports to be captured when connectivity is unreliable and synchronized when the connection returns.

Reported incidents are processed to understand their severity and priority, helping emergency teams focus on the most urgent needs. Operators can coordinate volunteers, resources and shelters from a centralized operations view and assign available help based on the incident requirements. By bringing incident reporting, AI-assisted understanding, prioritization and response coordination into one platform, ResQNet aims to reduce fragmented communication and help disaster-response teams make faster and more organized decisions.

What are the 3-4 Key Features?

1. Offline-First Emergency Reporting

Capture critical incident reports even when connectivity is unreliable and synchronize them when the connection returns.

2. AI-Assisted Incident Intelligence

Use AI to understand reported incidents and assist emergency teams in identifying their severity and priority.

3. Volunteer & Resource Coordination

Coordinate volunteers, their availability and required resources with the needs of reported incidents.

4. Unified Emergency Operations

Provide emergency teams with a centralized view of incidents, volunteers, resources, locations and shelters for coordinated response.

What makes it different or innovative?

- ResQNet focuses on the complete incident-to-response workflow rather than only collecting reports or displaying alerts. Its combination of offline-first reporting, AI-assisted incident prioritization, location-based information and volunteer/resource coordination helps turn scattered emergency information into coordinated action.

TECH STACK & ARCHITECTURE

Tech Stack

- **Frontend:** Next.js 15, React 19, TypeScript, Tailwind CSS, shadcn/ui
- **Backend:** FastAPI (Python), REST APIs, WebSockets
- **Database & Infra:** PostgreSQL, Redis, Docker / Docker Compose
- **AI & Maps:** LLM-based Incident Intelligence, Leaflet / React Leaflet

APIs/Libraries

- REST API
- WebSocket
- React Hook Form
- Zod
- Axios
- Leaflet / React Leaflet
- shadcn/ui

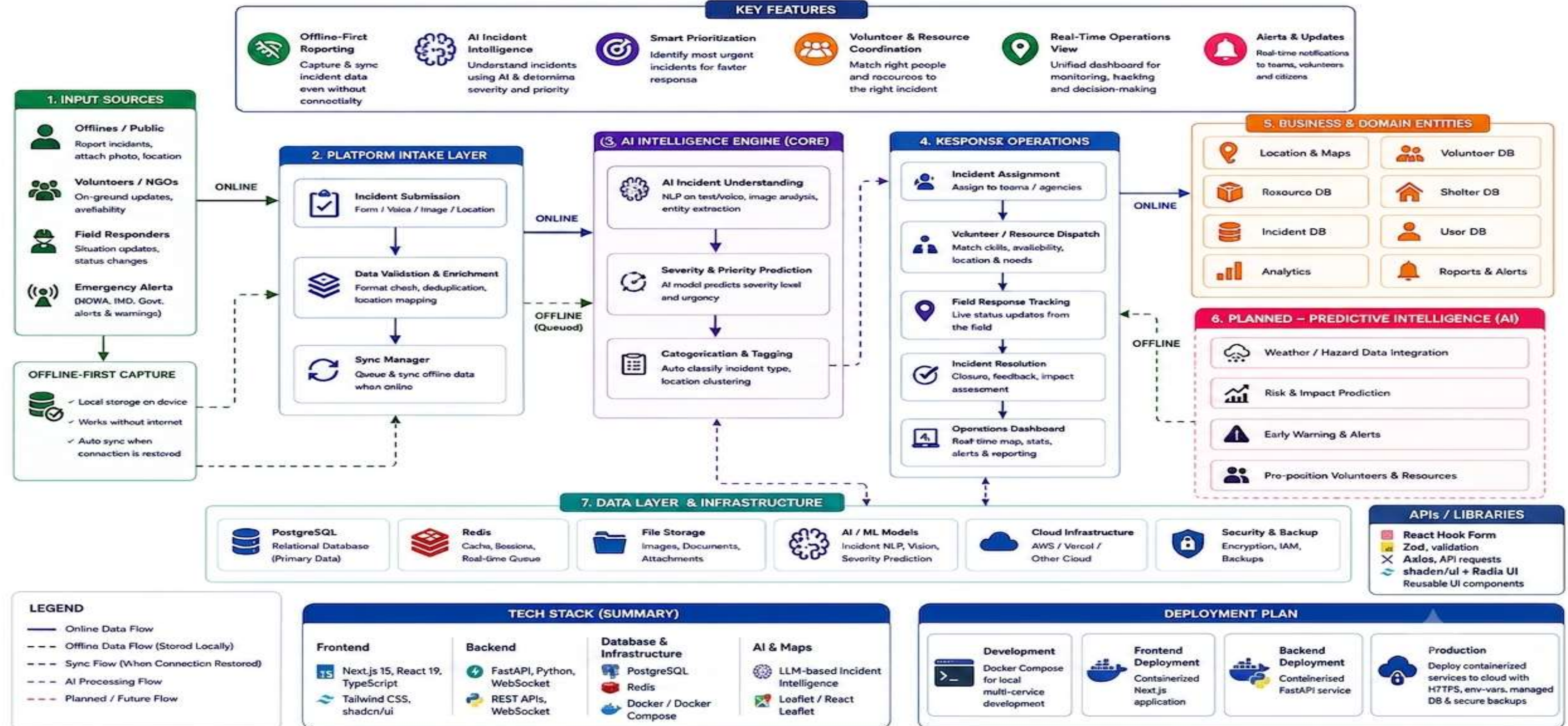
Deployment Plan

- **Development:** Docker Compose multi-service environment
- **Frontend:** Cloud deployment
- **Backend:** Containerized FastAPI service
- **Database:** Managed PostgreSQL + Redis
- **Production:** HTTPS, environment isolation and secure backups

ARCHITECTURE DIAGRAM

RESQNET - BLOCK DIAGRAM

Offline-First • AI-Powered • Real-Time Disaster Response Intelligence & Coordination



THANK
YOU

